Modeling of Electro-Mechanical Systems

(机电综合系统的建模)

Main Contents of Course (课程主要内容)

Mechanical (机械)

Electrical (电路)

Electro-Mechanical (机–电)

Fluid (流体)

Thermal (热系统)

Dynamic Modeling (动态建模)

Model structure (模型结构)

System Identification(系统辨识)

Parameter estimation (参数估计)

Control System Design(控制系统设计)

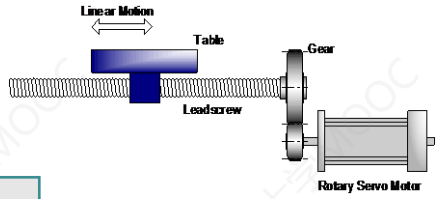
System Analysis (系统分析)

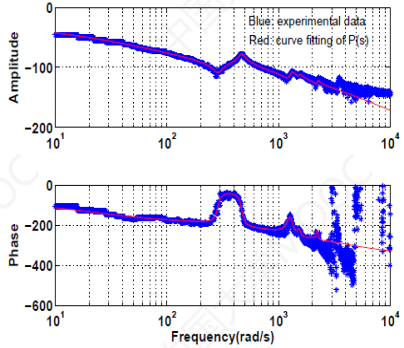
Root-locus design (根轨迹设计)

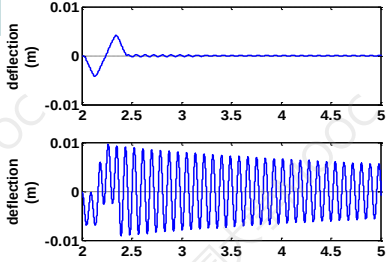
Pole placement design (极点配置)

Design limitations (性能局限)

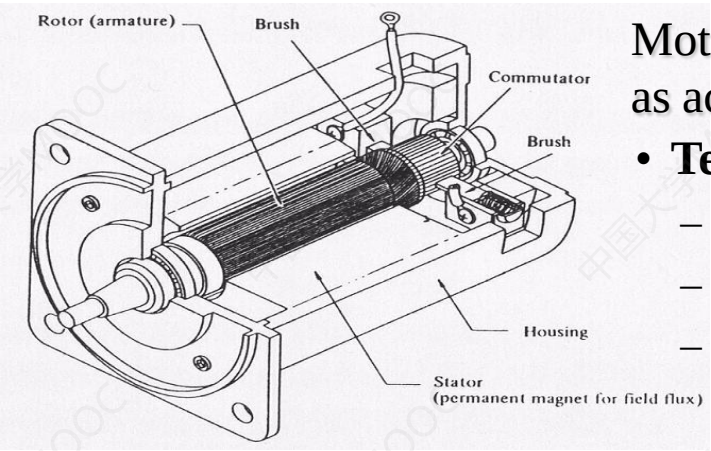
Architectural issues (控制结构)

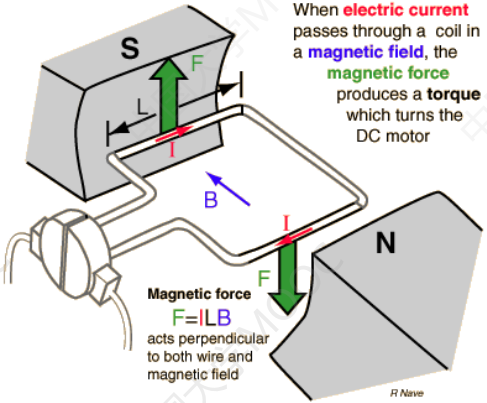






DC Motors





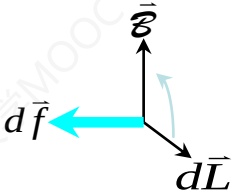
Motors are actuation devices (actuators) generating ***torque*** as actuation.**(电机是产生力矩作为驱动的执行器)**

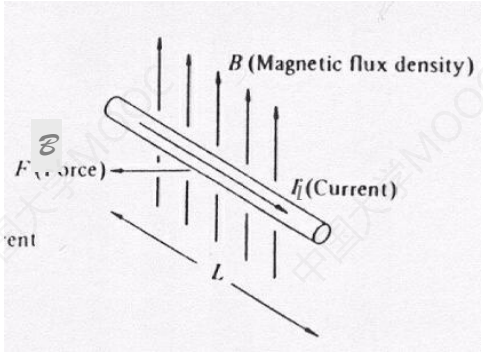
- **Terminology (术语)**
  - ***Rotor* (转子)** : the rotating part of the motor. **(电机的转动部分)**
  - ***Stator* (定子):** the stationary part of the motor. **(电机的固定部分)**
  - ***Field System* (励磁系统)** : the part of the motor that provides the magnetic flux**. (电机中提供磁通量的部分)**
  - ***Armature* (电枢)** : the part of the motor which carries current that interacts with the magnetic flux to produce torque. **(载有电流并与磁场交互作用产生力矩的电机部分)**
  - ***Brushes*(电刷)** : the part of the electrical circuit through which the current is supplied to the armature.**(使电流作用到电枢的电路部分)**
  - ***Commutator*(换向器)** : the part of the rotor that is in contact with the brushes.**(与电刷连接的转子部分)**

DC Motors — Principles of Operation

- **Torque Generation (产生的力矩)**

  Needs three elements: (所需因素)
  - Magnetic field (磁场)
  - Conductor (导体)
  - Current (电流)





➢Force will act on a conductor in a magnetic field with current flowing through the conductor: (磁场中通电导体产生的力)

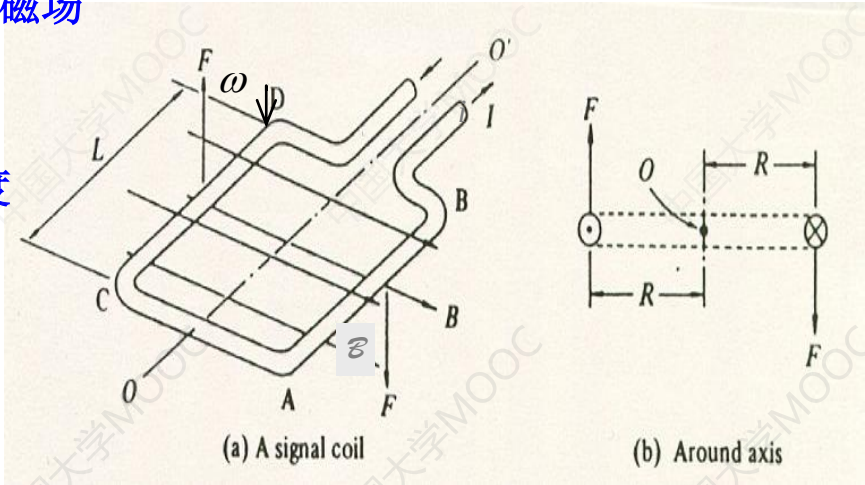
$$d\vec{f} = i_a \cdot d\vec{L} \times \vec{\mathcal{B}}$$

➢Integrate over the entire length: (在整个长度上积分)

$$f = i_a LB$$

➢Total torque generated: (产生的总力矩)

$$\tau_{\text{Coil}} = 2f.R$$
$$= 2i_a LB.R = 2BLR.i_a$$

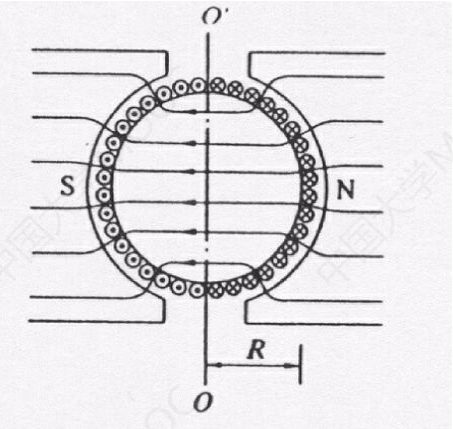


DC Motors — Principles of Operation

➢Let N be the number of coils in the motor. The total torque generated from the N coils is: (N匝线圈产生的总力矩)

$$\tau_m = N \cdot (2 \cdot \underbrace{i_a \cdot \mathcal{B} \cdot L}_{F} \cdot R)$$

$$= \underbrace{N \cdot (2 \cdot \mathcal{B} \cdot L \cdot R)}_{K_T} \cdot i_a$$



➢Define *Torque Constant* of the motor (定义电机的力矩常数)

$$K_T = 2 \cdot N \cdot \mathcal{B} \cdot L \cdot R \quad [\mathrm{Nm/A}]$$

➢The torque generated by a DC motor is proportional to the armature current i_a : (电机产生的力矩正比于电枢电流)

$$\tau_m = K_T \cdot i_a$$

For a DC motor, it is desirable to have a large K_T. However, size and other physical limitations often limits the achievable K_T (直流电机期望有大的K_T,但是，尺寸和其他物理制约经常限制了可取得的K_T值)

Large K_T :

– *Large (N, L, R).*

(N, L, R) is limited by the size and weight of the motor. **((N, L, R) 受电机尺寸和重量限制)**

– *Large $\mathcal{B}$:*

Need to understand the methods of generating flux **...(需要理解磁场产生的方法)**

DC Motors — Principles of Operation

- **Back-EMF Generation (反电动势)**

➢Electromotive force (EMF) is generated in a conductor moving in a magnetic field:(导体在磁场中运动产生电动势)

$$de_{emf} = (\vec{v} \times \vec{\mathcal{B}}) \cdot d\vec{L}$$

➢Integrate over the entire length L:(在整个长度上积分)

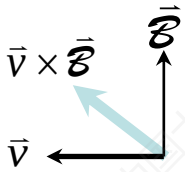
$$e_{emf} = vBL$$

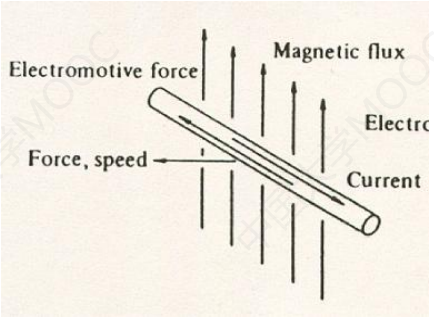
➢Since the N armature coils are in series, the total EMF is: (由于N个电枢线圈串联，总电动势是)

$$E_{emf} = 2N\underbrace{(R\omega)}_{v}\mathcal{B}L = \underbrace{2NR\mathcal{B}L}_{K_b}\omega$$

➢Define the *Back-EMF Constant* K_b :(定义反电动势常数)

$$K_b = 2 \cdot N \cdot R \cdot \mathcal{B} \cdot L \qquad [\mathrm{V}/(\mathrm{rad}/\mathrm{sec})]$$





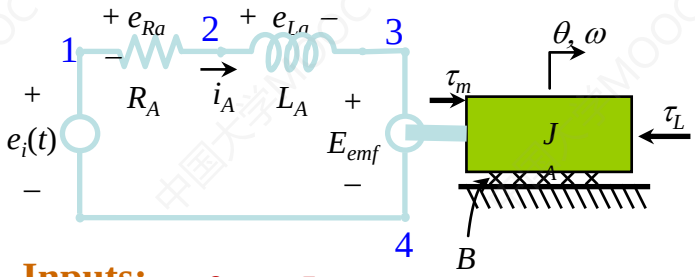
➢The Back-EMF due to the rotation of the motor armature (由于电机电枢转动产生的反电动势)

$$E_{emf} = K_b \cdot \omega$$

- is opposing the applied voltage (与施加的电压相反)
- is proportional to the angular speed ω of the motor (正比于电机转动角速度)
- note: $K_T = K_b$ is true only if SI unit is used ! (注意都采用标准单位时$K_T = K_b$ 成立)

DC Motors — Modeling

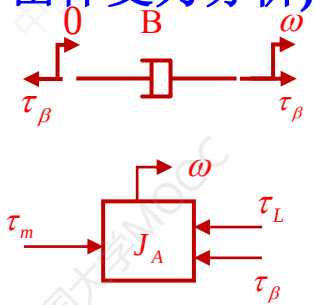
- **Schematic (原理图)**



Inputs: e_i, τ_L

Outputs: $y = \theta$

- **FBD (自由体受力分析)**



- **Element Equations (元件方程)**
  - **Electrical Subsystem (电路子系统)**

$$e_{12} = R_A i_A \qquad (1)$$

$$e_{23} = L_A \frac{di_A}{dt} \qquad (2)$$

  - **Mechanical Subsystem (机械子系统):**

$$J_A \dot{\omega} = \tau_m - \tau_L - \tau_B \qquad (3)$$

$$\tau_B = B(\omega - 0) = B\omega \qquad (4)$$

- **Interconnection Laws (互联关系)**

$$e_{12} + e_{23} + \underbrace{e_{34}}_{=E_{emf}} + \underbrace{e_{41}}_{=-e_i} = 0 \qquad (5)$$

$$\begin{cases} \tau_m = K_T i_A & (6) \\ E_{emf} = K_b \omega & (7) \end{cases}$$

- **Unknowns (未知变量)**

$$e_{12},\ i_A,\ e_{23},\ \theta = y,\ \tau_m,\ \tau_B,\ E_{emf}$$

DC Motors — Modeling

•**Derive I/O Model (I/O模型):**

$I_A(s) = \mathcal{L}[i_A], \quad E_i(s) = \mathcal{L}[e_i(t)]$

$E_{emf}(s) = \mathcal{L}[e_{emf}], \quad T_L(s) = \mathcal{L}[\tau_L(t)]$

$T_m(s) = \mathcal{L}[\tau_m]$

$$\begin{cases} R_A I_A(s) + L_A s I_A(s) + E_{emf}(s) = E_i(s) \\ J_A s^2 \theta(s) + Bs\theta(s) = T_m(s) - T_L(s) \\ T_m(s) = K_T I_A(s) \\ E_{emf}(s) = K_b s\theta(s) \end{cases}$$

$$\begin{cases} I_A(s) = \dfrac{1}{(L_A s + R_A)}\left[E_i(s) - E_{emf}(s)\right] \\ \Omega(s) = s\theta(s) = \dfrac{1}{J_A s + B}\left[T_m(s) - T_L(s)\right] \\ T_m(s) = K_T I_A(s) \\ E_{emf}(s) = K_b s\theta(s) = K_b \Omega(s) \end{cases}$$

$$\Rightarrow (J_A s + B)\Omega(s) = K_T \frac{1}{L_A s + R_A}\left[E_i(s) - K_b \Omega(s)\right] - T_L(s)$$

$$\Rightarrow (J_A s + B)(L_A s + R_A)\Omega(s) = K_T E_i(s) - K_T K_b \Omega(s) - (L_A s + R_A) T_L(s)$$

$$\{J_A L_A s^2 + (L_A B + R_A J_A)s + BR_A + K_T K_b\}\Omega(s) = K_T E_i(s) - (L_A s + R_A) T_L(s)$$

•**I/O Model from $e_i(t)$ and t_L to angular speed ω (从输入电压和负载力矩到角速度的I/O模型):**

$$\frac{L_A J_A}{K_T}\ddot{\omega} + \left(\frac{L_A B}{K_T} + \frac{R_A J_A}{K_T}\right)\dot{\omega} + \left(\frac{R_A B}{K_T} + K_b\right)\omega = e_i(t) - \frac{(L_A \dot{\tau}_L + R_A \tau_L)}{K_T}$$

•**I/O Model from $e_i(t)$ and t_L to angular position θ (从输入电压和负载力矩到角度的I/O模型):**

$$\frac{L_A J_A}{K_T}\dddot{\theta} + \left(\frac{L_A B}{K_T} + \frac{R_A J_A}{K_T}\right)\ddot{\theta} + \left(\frac{R_A B}{K_T} + K_b\right)\dot{\theta} = e_i(t) - \frac{(L_A \dot{\tau}_L + R_A \tau_L)}{K_T}$$

DC Motors — Modeling

- **Transfer Functions (传递函数):**

$$\Omega(s)=\overbrace{\frac{K_T}{J_AL_As^2+(BL_A+J_AR_A)s+BR_A+K_TK_b}}^{G_{\Omega E}}\cdot E_i(s)$$

$$+\underbrace{\frac{-(L_As+R_A)}{L_AJ_As^2+(L_AB+R_AJ_A)s+(R_AB+K_bK_T)}}_{G_{\Omega T}}\cdot T_L(s)$$

$$\Theta(s)=\frac{1}{s}G_{\Omega E}\cdot E_i(s)+\frac{1}{s}G_{\Omega T}\cdot T_L(s)$$

*Q: Let the load torque be zero (**No Load**), what is the steady state speed (**No-Load Speed**) of the motor for a constant input voltage V ?* (电机在负载力矩为零和常值输入电压下的稳态速度是多少)

$$e_i(t)=V=\text{constant}$$

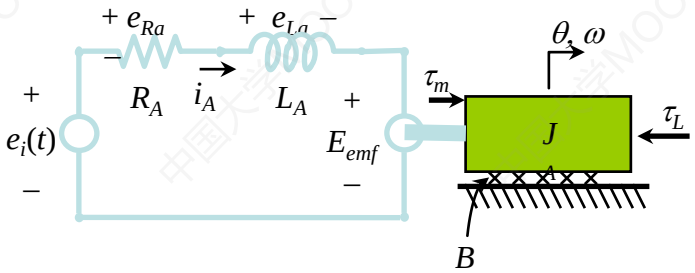
$$\Omega(s)=G_{\Omega E}E_i(s)=G_{\Omega E}\frac{V}{s}\Rightarrow\omega(t)\to\omega_{ss}=G_{\Omega E}(0)V=\frac{K_T}{BR_A+K_TK_b}V$$

*Q:Let the load torque τ_L = T, what is **the steady state speed of the motor** for a constant input voltage V ?*
(电机在常值负载力矩和常值输入电压下的稳态速度是多少)

$$\omega(t)\to\omega_{ss}=G_{\Omega E}(0)V+G_{\Omega T}(0)T=\frac{K_T}{BR_A+K_TK_b}V-\frac{R_A}{BR_A+K_TK_b}T$$

Block Diagram Representation of DC Motors

- **Schematic** (原理图)



- **Governing Equations** (主导方程)

$$L_A \frac{d}{dt} i_A + R_A i_A + E_{emf} = e_i(t) \quad \text{(Electrical (电路))}$$

$$J_A \dot{\omega} + B\omega = \tau_m - \tau_L \quad \text{(Mechanical (机械))}$$

$$\left.\begin{aligned} \tau_m &= K_T \cdot i_A \\ E_{emf} &= K_b \cdot \omega \end{aligned}\right\} \quad \text{(EM Coupling (机电耦合))}$$

- **EOM in *s*-Domain** (*s*域的运动方程)

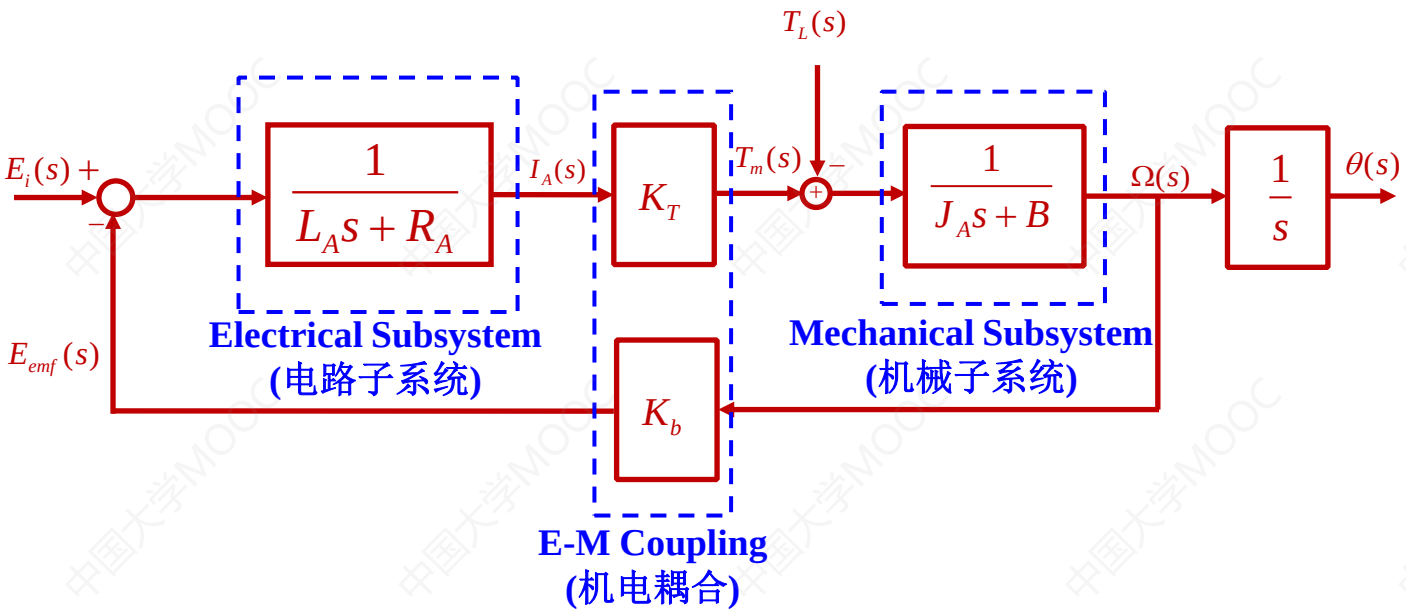
$$\Rightarrow I_A(s) = \frac{1}{L_A s + R_A}\left[E_i(s) - E_{emf}(s)\right]$$

$$\Omega(s) = \frac{1}{J_A s + B}\left[T_m(s) - T_L(s)\right]$$

$$T_m(s) = K_T I_A(s)$$

$$E_{emf}(s) = K_b \Omega(s)$$

Block Diagram Representation of DC Motors



Example

(A) Given the following specification of a DC motor and assume there is no load, find its transfer function from input voltage to motor angular speed (已知直流电机的下列参数，并假设没有负载，找出从输入电压到电机角速度的传递函数)

L_A = 2 mH

R_A = 10 Ω

K_T = 0.06 Nm/A

J_A = 5×10^{-6} Kg m^2

B = 3 ×10^{-6} Nm/(rad/sec)

$$G_{\Omega E}(s)=\frac{0.06}{2\times10^{-3}\times5\times10^{-6}s^2+(3\times10^{-6}\times2\times10^{-3}+10\times5\times10^{-6})s+(10\times3\times10^{-6}+0.06^2)}$$

$$=\frac{6\times10^{-6}}{s^2+5000s+3.63\times10^5}$$

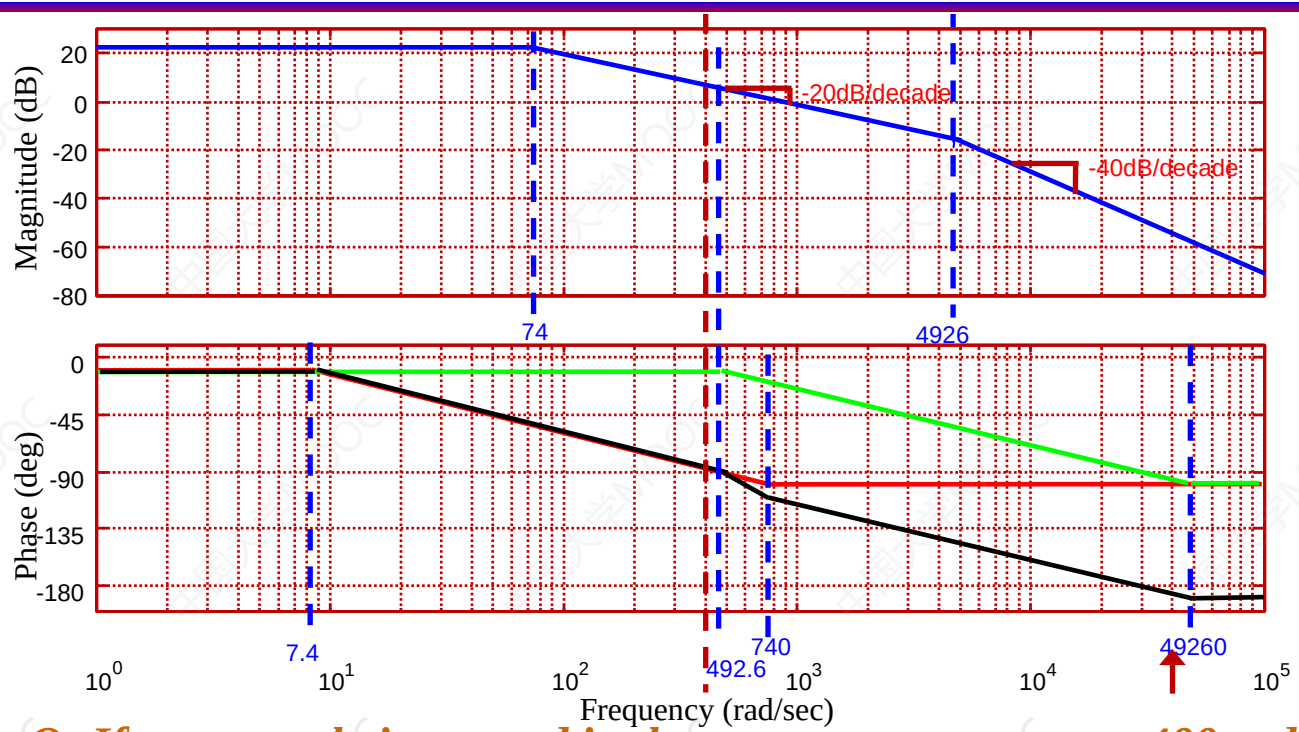
(B) Find the poles of the transfer function (找出传递函数的极点)

$$s^2+5000s+3.63\times10^5=0$$

$$\Rightarrow\begin{cases}p_1=-74\\p_2=-4926\end{cases}$$

(C) Plot the Bode diagram of the transfer function (画出传递函数伯德图)

Example



$$G_{\Omega E}(s) == \frac{6\times10^6}{(s-p_1)(s-p_2)}$$

$$=16.53\times\frac{1}{\frac{1}{74}s+1}\cdot\frac{1}{\frac{1}{4926}s+1}$$

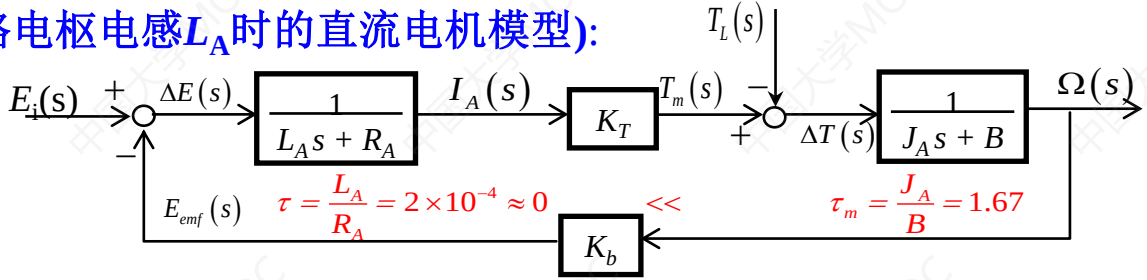
Q: If we are only interested in the system response up to 400 rad/sec, can we simplify our model ? How would you simplify the model ? **(如果我们只对400rad/sec频宽以内的系统响应感兴趣，如何简化模型)**

$$G_{\Omega E}(s)\approx16.53\times\frac{1}{\frac{1}{74}s+1}\quad \text{for } \omega<400 \text{ rad/s}$$

Model Reduction

- **Neglect Electrical Dynamics** **(忽略电路动力学)**

Derive the model for the DC motor, if the armature inductance L_A is neglected **(推导出忽略电枢电感L_A时的直流电机模型)**:



$$\Omega(s) = \frac{K_T}{\cancel{L_A J_A s^2} + (\cancel{BL_A} + R_A J_A)s + (R_A B + K_b K_T)} \cdot E_i(s) - \frac{\cancel{L_A} s + R_A}{\cancel{L_A J_A s^2} + (\cancel{BL_A} + R_A J_A)s + (R_A B + K_b K_T)} \cdot T_L(s)$$

$$\Rightarrow \quad \Omega(s) = \underbrace{\frac{K_T}{R_A J_A s + (R_A B + K_b K_t)}}_{G_{\Omega E}(s)} \cdot E_i(s) - \underbrace{\frac{R_A}{R_A J_A s + (R_B B + K_b K_T)}}_{G_{\Omega T}(s)} \cdot T_L(s)$$

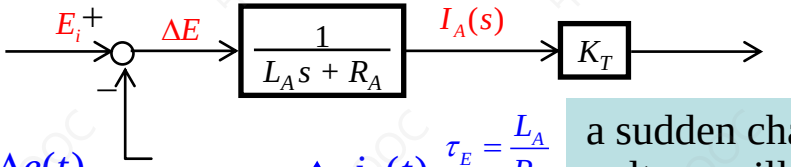
Simplified to: First-order system
(简化成一阶模型)

By neglecting the effect of armature inductance, we reduced the order of the model from two to one when the output is the motor speed ! (忽略电枢电感，以电机速度为输出，模型可以从二阶降到一阶)

Model Reduction

Q: Physically, what do we mean by neglecting armature inductance ? (物理上忽略电枢电感意味着什么)

- It takes no time for the current to reach its steady state value when there is a step change in input voltage (当输入电压阶跃改变时，电流达到稳态值不需要响应时间)
- This is equivalent to have direct control of the motor current (等效于电机电流的直接控制)



Mathematically (动态方程):

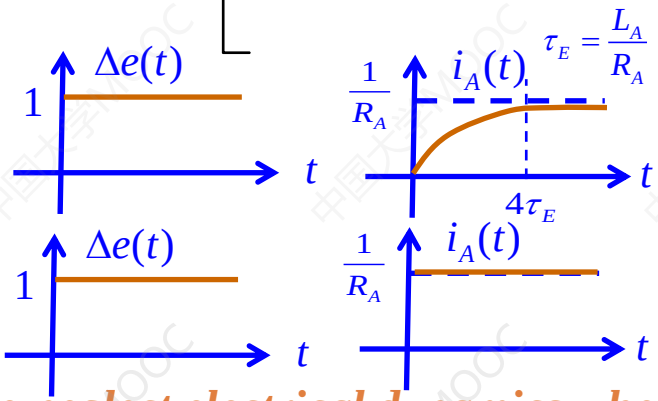
$$L_A \frac{d}{dt} i_A + R_A i_A + E_{emf} = e_i(t)$$

$$J_A \dot{\omega} + B\omega = T_m - T_L$$

$$\left.\begin{aligned} T_m &= K_T \cdot i_A \\ E_{emf} &= K_b \cdot \omega \end{aligned}\right\}$$

Reality (实际)

Neglected (忽略L_A)



a sudden change in input voltage will result in a sudden change in the armature current, which in turns will result in a sudden change in the motor torque output (突变的输入电压变化引起突变的电枢电流，从而引起突变的电机力矩输出).

Q: Practically, when can we neglect electrical dynamics when modeling motors?

(实际对电机建模时什么时候可以忽略电路动力学呢)

$$\tau_E = \frac{L_A}{R_A} << \tau_m = \frac{J_A}{B}$$

Summary

- **DC Motors (直流电机)**
  - Principles of Operation **(工作原理)**
  - Modeling (EOM) **(运动方程建模)**
- **Block Diagram Representations (方框图表示)**
  - A Convenient Graphic Representations of Interconnections among various subsystems of a complex system described by Equations in *s* – Domain **(由s域方程描述的复杂系统的各种子系统之间互联关系的方便图形表示方法)**
  - Block Diagram Representation of DC Motors **(直流电机的方框图表示)**